

```
import pandas as pd
```

```
df=pd.read_csv('/content/gender_submission.csv')
df
```

	PassengerId	Survived
0	892	0
1	893	1
2	894	0
3	895	0
4	896	1
...	...	...
413	1305	0
414	1306	1
415	1307	0
416	1308	0
417	1309	0

418 rows × 2 columns

```
x=df.iloc[:, :-1]
y=df.iloc[:, -1]
```

```
import numpy as np
```

```
x=np.array(x).reshape(-1,1)
y=np.array(y).reshape(-1,1)
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42
)
print("Training samples:",x_train.shape)
print("Testing samples:",x_test.shape)
```

```
Training samples: (334, 1)
Testing samples: (84, 1)
```

```
from sklearn.svm import SVC
```

```
svm_linear=SVC(kernel='linear',class_weight='balanced',C=0.01)
```

```
svm_linear.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the s
y = column_or_1d(y, warn=True)
```

```
▼ SVC ⓘ ?
SVC(C=0.01, class_weight='balanced', kernel='linear')
```

```
y_pred=svm_linear.predict(x_test)
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,recall_score,f1_score
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,recall_score,f1_score
precision_score_val = precision_score(y_test,y_pred)
print("Precision:",precision_score_val)
```

```
Precision: 0.42857142857142855
```

```
accuracy=svm_linear.score(x_test,y_test)
print("Accuracy:",accuracy)
```

```
Accuracy: 0.5238095238095238
```

```
from sklearn.metrics import confusion_matrix
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
print("Confusion Matrix:")
print("TP FP")
print("FN TN")
print(cm)
```

```
Confusion Matrix:
TP FP
FN TN
[[18 16]
 [24 26]]
```

```
recall = recall_score(y_test, y_pred, zero_division=0)
print("Recall:",recall)
```

```
Recall: 0.5294117647058824
```

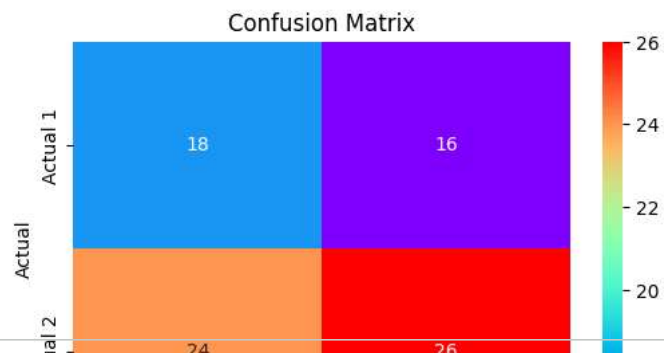
```
f1=f1_score(y_test, y_pred, zero_division=0)
print("F1 Score:",f1)
```

```
F1 Score: 0.47368421052631576
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm,annot=True,fmt='d',cmap='rainbow',
            xticklabels=['Predicted 1','Predicted 2'],
            yticklabels=['Actual 1','Actual 2'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
svm_linear=SVC(kernel='poly',class_weight='balanced',C=0.01)
svm_linear.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the s
y = column_or_1d(y, warn=True)
SVC
SVC(C=0.01, class_weight='balanced', kernel='poly')
```

```
y_pred=svm_linear.predict(x_test)
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,recall_score,f1_score
```

```
accuracy=accuracy_score(y_test,y_pred)
```

```
print("Accuracy",accuracy)
```

```
Accuracy 0.5238095238095238
```

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
print('Confusion Matrix')
print(' TP FP')
print(' FN TN')
print(cm)
```

```
Confusion Matrix
TP FP
FN TN
[[20 14]
 [26 24]]
```

```
precision=precision_score(y_test,y_pred,zero_division=0)
recall=recall_score(y_test,y_pred,zero_division=0)
f1=f1_score(y_test,y_pred,zero_division=0)
print("Precision",precision)
print("Recall",recall)
print("F1 Score",f1)
```

```
Precision 0.43478260869565216
Recall 0.5882352941176471
F1 Score 0.5
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
```

```
plt.figure(figsize=(8,6))
sns.heatmap(cm,annot=True,fmt='d',cmap='rainbow',xticklabels=['predicted 1','predicted 0'],yticklabels=['actual 1','actual 0'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

